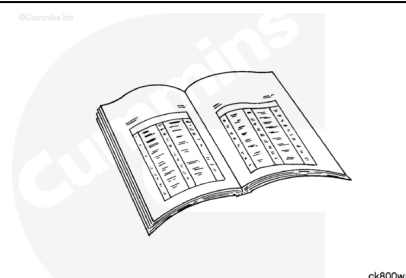


Trainings ▾

Preparatory Steps

Automotive, Industrial with CM570 and Power Generation without CM570

- Disconnect the air pipe from the charge air cooler to the air inlet connection.
- Disconnect the boost sensor wire.
- Remove the four-step wastegate controller wires, hose, mounting capscrews, and controller.



Automotive with CM870

- Remove the charge air cooler piping. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-027.html>)
- Remove the EGR mixer. Refer to Procedure 011-021 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-021-tr.html>)
- Disconnect the engine wiring harness from the intake manifold pressure sensor and intake manifold air temperature sensor.

Automotive With CM871

Note : Brush away any loose dirt from around the area of the air handling connections to avoid contamination of the interior of the engine.

- Remove the charge air piping. Use protective caps from the Air Handling Clean Care Kit to cover the connection points.
- Intake manifold pressure temperature sensor. For ISX and ISM engines, see the following procedure in the CM871 and CM876 Troubleshooting and Repair Manual Electronic Control System ISX and ISM Engines, Bulletin 4021560. Refer to Procedure 019-370 in Section 19. (</qs3/pubsys2/xml/en/procedures/16/16-019-370.html>)

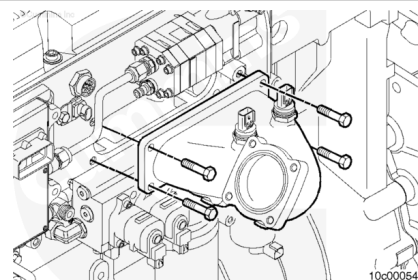
- Remove the air compressor supply line. Refer to Procedure 012-014 in Section 12. (</qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html>)
- Front EGR Connection tube and EGR venturi. Refer to Procedure 011-025 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html>)
- EGR differential pressure sensor. For ISX and ISM engines, see the following procedure in the Electronic Control System CM871 and CM876 Signature, ISX and QSX15 Engines, Bulletin 4021560. Refer to Procedure 019-370 in Section 19. (</qs3/pubsys2/xml/en/procedures/16/16-019-370.html>)
- EGR differential pressure sensing tubes. Refer to Procedure 011-027 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-027-tr.html>)

Remove

Automotive with CM870

Remove the four capscrews holding the air intake connection to the cylinder head.

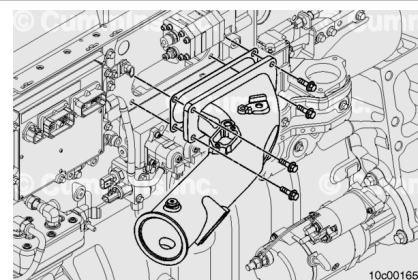
Remove and discard the gasket.



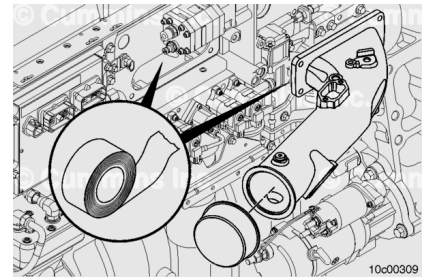
Automotive With CM871

Remove the four capscrews that hold the air intake connection to the cylinder head.

Remove and discard the gasket.



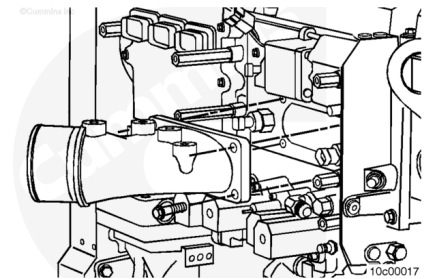
Cover all open points on the air intake connection and cylinder head with caps and heavy tape from the Air Handling Clean Care Kit or heavy tape.



Automotive, Industrial with CM570 and Power Generation without CM570

Remove the four mounting capscrews, air inlet connection, and gasket.

Remove and discard the gasket.

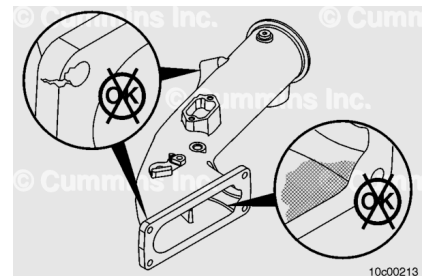


Clean and Inspect for Reuse

All Applications

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



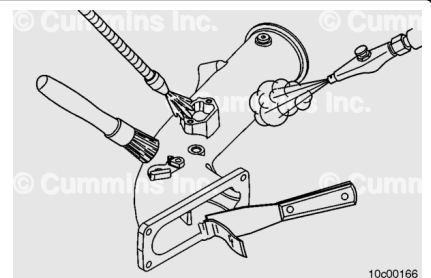
Inspect the air intake connection for the amount of carbon on the inside of the air intake connection.

Clean the air intake connection. Saturate the inside of the air intake with safety solvent. Mineral based solvents can be used if expected results with the safety solvent are **not** acceptable.

Clean the debris with a non-abrasive brush. Dry the air intake connection with compressed air.

Inspect for cracks or damage threads. Replace if necessary.

Inspect the cylinder head air intake area.



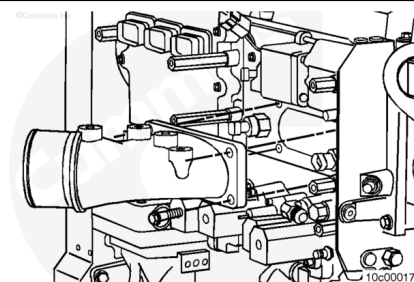
Some soot build up inside the cylinder head is normal and will **not** affect engine performance. Clean the air intake to cylinder head surface with Scotch Brite™ 7448 abrasive pad or a gasket scraper. Do **not** use power tools combined with an abrasive pad to clean the gasket surfaces.

Install

Automotive, Industrial with CM570 and Power Generation without CM570

Install the four mounting capscrews, air inlet connection, and new gasket.

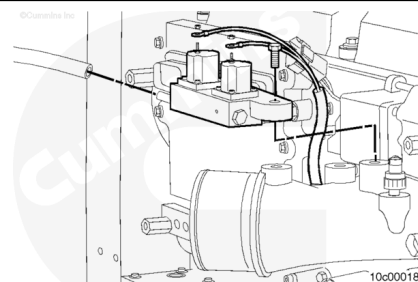
Torque Value: 45 n•m [33 ft-lb]



Install the four-step wastegate controller, mounting capscrews, hose, and controller wires.

Connect the boost sensor wire.

Connect the air pipe from the charge air cooler to the air inlet connection.

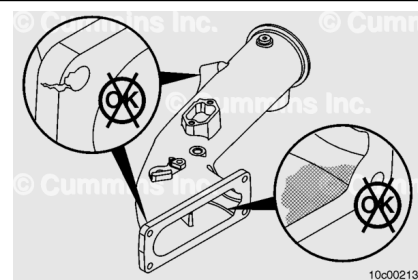


Automotive with CM870

Install a new gasket between the cylinder head and the air intake connection.

Install and tighten the four capscrews.

Torque Value: 45 n•m [33 ft-lb]

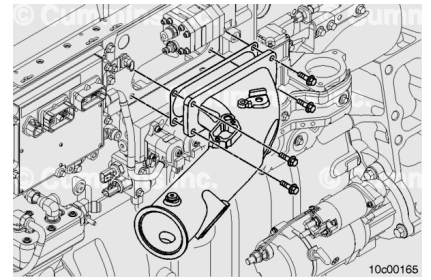


Automotive With CM871

Install a new gasket between the cylinder head and the air intake connection.

Install and tighten the four capscrews.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

Automotive, Industrial with CM570 and Power Generation without CM570

- Connect the air pipe from the charge air cooler to the air inlet connection.
- Connect the boost sensor wire.
- Install the four-step wastegate controller wires, hose, mounting capscrews, and controller.



Automotive with CM870

- Install the four-step wastegate controller, mounting capscrews, hose, and controller wires.
- Connect the boost sensor wire.
- Connect the air pipe from the charger air cooler to the air inlet connection.
- Connect the engine wiring harness to the intake manifold pressure sensor and intake manifold air temperature sensor.
- Install the EGR mixer. Refer to Procedure 011-021 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-021-tr.html>)
- Install the charge air cooler piping. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-027.html>)
- Start and operate the engine. Check for air leaks.

Automotive With CM871

- Install the front EGR connection tube and EGR venturi. Refer to Procedure 011-025 in Section 11.

(/qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html)

- Install the air compressor supply line. Refer to Procedure 012-014 in Section 12.

(/qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html)

- Install the intake manifold pressure/temperature sensor. For ISX and ISM engines, see the following procedure in the CM871 and CM875 Troubleshooting and Repair Electronic Control System ISX and ISM Engines, Bulletin 4021560. Refer to Procedure 019-370 in Section 19.

(/qs3/pubsys2/xml/en/procedures/16/16-019-370.html)

- Install the charge air cooler piping. Refer to Procedure 010-027 in Section 10.

(/qs3/pubsys2/xml/en/procedures/10/10-010-027.html)

- Operate the engine and check for air leaks.

If damage resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14. (/qs3/pubsys2/xml/en/procedures/101/101-014-013.html)